Panlin Li (Direct Ph.D. Application)

High-Dimensional Statistical Learning | Data Modeling

2004lryan@gmail.com · (+86) 134-6841-2787

https://2004lryan.github.io/

Member, China Society for Industrial and Applied Mathematics · Member, Shandong Young Photographers' Association · Member, Qingdao Young Photographers' Association

Education

Xinjiang Agricultural University (School of Mathematics & Physics)	Mathematics & Applied Mathematics · B.S. (Top 2.5%)	Advisor: Prof. Hua Huang	Sep. 2023 – Jul. 2027

- Honors: Outstanding Volunteer (9 times), Outstanding Team, and "Child Safety Guardian Plan" Outstanding Team (team lead) —Central Committee of the Communist Youth League; Outstanding Student, Innovation & Entrepreneurship Star (two consecutive years), and Science & Innovation Pioneer Scholarship —Xinjiang Agricultural University.
- Key outcomes: led/participated in 3 national-level, 4 regional-level, and 7 university-level projects; first-authored 8 academic papers (plus 3 SCI Q1 Top under review: CEA IF 8.9, Food Chemistry IF 9.8, Information Fusion IF 15.5); granted 4 utility-model patents and 4 software copyrights; received 5 national-level and 6 provincial-level competition awards; project recognized by Huawei Research; volunteer work covered by *China Daily*, *People's Daily*, and other media outlets.

Research & Projects

National Statistical Science Research Project · Major Program · Under Application **Multimodal High-Dimensional Statistical Learning for Cotton Yield and Climate Risk** **2026 – Present**

- Responsibilities: lead-authored the National Major Program proposal and a 7,000-word survey under advisor guidance, mapping the frontier of cotton-yield prediction and climate attribution; proposed a heterogeneous-multimodal tensor-sparse fusion model jointly modeling meteorological / soil / remote-sensing / phenological signals; designed a spatio-temporal doubly-debiased Lasso + generalized random forest causal-identification framework that separates the marginal effect of climate shocks from confounding bias; planned a multi-stream INSPECT online change-point detector coupled with PCMCI for field-level early warning; completed preliminary method derivation and pre-experiment data preparation.
- Key outcomes: proposal submitted and currently under review.

Xinjiang Natural Science Foundation · General Program · Participant **Functional Statistical Modeling via Spatial-Spectral Feature Fusion of Hyperspectral Imaging for Intelligent Detection of Agricultural Products** **2026 – Present**

- Responsibilities: curated public multi-cultivar HSI/NIR datasets; performed spatial-spectral preprocessing, SNV/MSC correction, and multi-basis (B-spline / wavelet / Fourier) functional representation; addressed feature alignment, mechanism priors, and sparse multi-task respectively through three algorithms — *functional contrastive sparse multi-task learning*, *mechanism-embedded post-harvest modeling*, and *pesticide-residue spectral screening*; lead-authored the three corresponding papers.
- Key outcomes: three first-author drafts completed (under advisor review).

National Innovation Project · Ongoing · Participant **Apple Origin Traceability Model via Machine Learning & Transfer Learning** **2025 – Present**

- Responsibilities: lead-authored the project proposal; led cross-year apple HSI and image data collection across Xinjiang/Shandong/Gansu with unified SSC and origin labels; built a 9-domain benchmark with an origin × year double-layered leakage-controlled evaluation protocol that removes the over-optimism of single-split evaluation; ran a PLS-DA / SVM / 1D-CNN / Transformer baseline comparison and cross-year generalization study; led paper writing and submission.
- Key outcomes: first-author submission to SCI Q1 Top journal *Computers and Electronics in Agriculture* (IF: 8.9; under review).

National Project · Completed · PI **Small-Sample Time-Series Forecasting in Energy Economics** **2024 – 2025**

- Responsibilities: lead-authored the proposal and coordinated project execution; re-engineered the non-homogeneous grey model via *conformable fractional accumulation* (CFA) into KR-NGM(1,1) with a closed-form solution for the kernel-regularized objective, relaxing the periodicity/monotonicity assumptions of classical grey models; benchmarked against GM(1,1) / DGM / FNGM / SVR on energy-consumption and GDP small-sample time series with error and robustness analyses; co-authored the paper.
- Key outcomes: project successfully completed; co-first-author publication in *Transactions on Computational and Applied Mathematics* (2024, 4: 137–147).

Regional Innovation Project · Ongoing · Participant **Hyperspectral Agricultural Product Detection via Functional Contrastive Sparse Multi-Task Learning** **2026 – Present**

- Responsibilities: lead-authored the regional Innovation Project proposal; on functional Hilbert spaces, built a unified FCPCA + sparse multi-task framework (variational substitution + block-coordinate descent + group-sparse proximal-gradient solver), integrating chemistry-anchored priors and image/spectrum/functional tri-modal alignment; consolidated 10 public datasets (30+ GB) of HSI/NIR/RGB with cross-cultivar *leave-one-class-out* validation; lead-authored the paper.
- Key outcomes: first-author submission to SCI Q1 Top journal *Information Fusion* (IF: 15.5; under review).

Regional Innovation Project · Excellent Completion · Participant **Apple SSC Prediction via Functional Data Analysis with Image & Spectral Feature Fusion** **2024 – 2026**

- Responsibilities: collected cross-origin apple HSI and image data with SSC labels in Xinjiang/Shandong/Gansu; used raw spectra and their 1st/2nd-order derivatives as functional variables with multi-basis representation and basis selection; extracted image chromaticity and texture features; implemented a *semi-functional partial linear regression* fusing image and spectrum with GCV-based estimation; co-wrote the paper.
- Key outcomes: fourth-author publication in *Spectroscopy and Spectral Analysis* (2025, 45(S1): 415–423).

Regional Innovation Project · Excellent Completion · PI **Wenmai Zhiyuan Hub — Rural Revitalization, Roots of Culture, Whispers of the Soul** 2024 – 2025
University Innovation Project · Excellent Completion · PI **Nonlinear Wave Solutions of High-Dimensional Nonlinear PDEs** 2024 – 2025
University Innovation Project · Excellent Completion · Participant **Zhitong Smart Cane — Intelligent Blind Cane Empowered by Huawei Cloud & MindSpore** 2025 – 2026
University Innovation Project · Excellent Completion · Participant **Tianshan Pastoral Song — Nomadic Dairy Lineage Revitalization Plan** 2025 – 2026
University Innovation Project · Pass · Participant **“Cure Arrival” — AI Pest & Disease Diagnosis System for Rural Revitalization** 2025 – 2026
University Innovation Project · Pass · Participant **Novel Traveling-Wave Solutions and Localized Excitations of Nonlinear PDEs** 2025 – 2026
University Innovation Project · Ongoing · PI **On the Rapid Detection of Agricultural Products via Spectral Transfer across Heterogeneous** 2026 – Present
University Innovation Project · Ongoing · Participant **Fruit & Vegetable Storage Quality Evolution Forecasting via Spectral Time-Series Modeling** 2026 – Present

Publications

- Li, Panlin; Zhang, Qingli; Jiao, Qitai; et al. *Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study*. Computers and Electronics in Agriculture. (SCI Q1 Top; IF: 8.9; Under Review)
- Li, Panlin; Li, Longjie; Liu, Ya; et al. *Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior*. Food Chemistry. (SCI Q1 Top; IF: 9.8; Under Review)
- Li, Panlin; Li, Yuchang; Huang, Hua; et al. *Functional Multi-Task Sparse Learning with Spectral-Spatial Information Fusion for Hyperspectral Food Quality Detection*. Information Fusion. (SCI Q1 Top; IF: 15.5; Under Review)
- Li, Panlin; Li, Jie; Alamusijiang Kadir; et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Li, Panlin; Bao, Fangchen; Zhang, Jingyi; et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- Li P, Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- Li P, Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- Li P, Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- Li P, Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004. doi: 10.1109/ICCECT64621.2025.11339538.
- Li P, Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- Rao W J, Li P L. *An improved kernel regularized nonhomogeneous grey model based on conformable fractional accumulation*. Transactions on Computational and Applied Mathematics, 2024, 4: 137–147. (Co-First Author)
- Zhang, Ning; Li, Panlin; Bao, Fangchen; Liu, Xiaoyu. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao, Qitai; Li, Panlin; Mei, Tao; et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- Huang, Hua; Wang, Mo; Liu, Ya; Li, Panlin; et al. *Prediction of Apple Soluble Solids Content Using a Functional Partial Linear Regression Model Fusing Image Color Features and Vis-NIR Spectra*. Spectroscopy and Spectral Analysis, 2025, 45(S1): 415–423.
- Xinjiang Agricultural University (Li, Panlin; Huang, Hua; Zhang, Ning; et al.). *Section-Connection Component for a White Cane*. Patent CN2024230029031, 2024-12-05. (Granted)
- Xinjiang Agricultural University (Li, Panlin; Huang, Hua; Zhang, Ning; et al.). *Portable Ultrasonic Detection Device*. Patent CN2024230026103, 2024-12-05. (Granted)
- Xinjiang Agricultural University (Li, Panlin; Huang, Hua; Zhang, Ning; et al.). *Pesticide Spraying Drone*. Patent CN20241226759, 2024-12-26. (Granted)
- Xinjiang Agricultural University (Li, Panlin; Huang, Hua; Zhang, Ning; et al.). *Foldable Anti-Collision Spraying Boom for Drones*. Patent CN20241226785, 2024-12-26. (Granted)
- Li, Panlin; Huang, Hua; Zhang, Ning; et al. *Intelligent White Cane Vision Recognition System V1.0*. Software Copyright No. 2025SR0470648, 2025-03-17.
- Li, Panlin; Huang, Hua; et al. *Smart Cane Data Collection & Monitoring System V1.0 (Huawei Cloud MindSpore)*. Software Copyright No. 2025SR0188010, 2025-01-27.
- Li, Panlin. *User Behavior Analysis and Personalized Recommendation System V1.0*. Software Copyright No. 2024SR1458187, 2024-09-30.
- Li, Panlin. *Big-Data-Driven Market Analysis and Forecasting System V1.0*. Software Copyright No. 2024SR1124927, 2024-08-05.

Competitions & Awards

- 2026 (16th) "Chia Tai Cup" National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2026.05
- 2025 (11th) National Undergraduate Statistical Modeling Contest · Undergraduate Third Prize *Team Lead* 2025.10
- 2025 (15th) "Chia Tai Cup" National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2025.05
- 2024 China International College Students' Innovation Competition · Bronze Award *Participant* 2024.10
- 2024 DataBee Cup National College Student Interview Survey Contest · Third Prize *Participant* 2024.10
- 2025 China International College Students' Innovation Competition (Xinjiang Division) · Bronze Award *Team Lead* 2025.08
- 2025 (14th) China Innovation & Entrepreneurship Competition (Xinjiang Division) · Growth Group Third Prize *Team Lead* 2025.10
- 2025 (11th) National Undergraduate Statistical Modeling Contest (Xinjiang Division) · Undergraduate First Prize *Team Lead* 2025.08
- 2024 National College Students Mathematical Modeling Contest (Xinjiang Division) · Undergraduate Second Prize *Team Lead* 2024.11

- 2024 (10th) National Undergraduate Statistical Modeling Contest (Xinjiang Division) · Second Prize *Team Lead* 2024.07
- 2024 Information Literacy Competition (Xinjiang Division) · Excellence Award *Team Lead* 2024.12

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics 2023.09 – 2027.07
Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University 2025.05 – 2026.05
Head, Volunteer Association, School of Mathematics & Physics, Xinjiang Agricultural University 2024.09 – 2025.11

- As Class Monitor, organized academic-style building and sports activities; led the class to receive the *Outstanding Class Collective* and *Sports Pioneer Class* honors at Xinjiang Agricultural University (AY 2024–2025).
- Founded and built the "Shield of the Future" university volunteer team (20 members) and joined the China Youth Volunteer Association; participated in 11 public-welfare programs hosted by the CYL Central Committee and CYVA, including teaching support and rural revitalization, with footprints across counties and cities in Shandong, Henan, and Xinjiang; the team's work was reported by *China Daily*, *People's Daily*, and other media outlets; received the title of National Outstanding Youth Volunteer Team, and was subsequently invited to join the School of Mathematics & Physics Volunteer Association.

Skills

- **Language:** TOEFL 5.5/6; first-authored multiple English SCI/IEEE papers; capable of independent English academic writing.
- **Professional:** high-dimensional statistical inference (debiased Lasso, double machine learning, generalized random forests), functional data analysis (functional PCA, function-on-function regression), tensor decomposition and multimodal fusion; proficient in Python (PyTorch / scikit-learn), capable of independent end-to-end modeling.
- **Tools:** LaTeX for academic writing, Git for version control, Lightroom / Final Cut Pro for imaging.

Self-Assessment

- I believe in "solving problems first, publishing papers second." Through 14 projects and 11 papers during my undergraduate years, I have walked through mathematics, statistics, machine learning, and real-world application scenarios — which has made me clearly aware that my bottleneck lies in methodology and theory. In the Ph.D. stage, I aim to deepen my work in high-dimensional statistics and functional modeling, and make methods truly usable in cotton fields, orchards, and rural communities. *Ad infinitum.*